

36.3 Proportional odds and its checks

Setting $\beta_1 = \dots = \beta_{c-1} = \beta$ in Eq. 36.2.2 turns $(c-1)p$ slopes into p . With the logistic link the result is the model that carries the name of its most conspicuous consequence.

Theorem 36.3.1 — The proportional odds model

Let

$$\log \frac{P(Y_i \leq r \mid \mathbf{x}_{(i)})}{P(Y_i > r \mid \mathbf{x}_{(i)})} = \theta_r - \mathbf{x}_{(i)}^\top \beta, \quad r = 1, \dots, c-1. \quad (36.3.1)$$

Then:

- (a) (*latent variable*) Eq. 36.3.1 holds exactly when there is a latent response $U_i = \mathbf{x}_{(i)}^\top \beta + \varepsilon_i$, with ε_i independent of $\mathbf{x}_{(i)}$ and standard logistic, and $Y_i = r$ precisely when $\theta_{r-1} < U_i \leq \theta_r$, where $-\infty = \theta_0 < \theta_1 < \dots < \theta_{c-1} < \theta_c = \infty$. Replacing the logistic distribution by any continuous, strictly increasing G gives the general cumulative link model of Definition 36.2.1 with common slopes;
- (b) (*collapsing*) let $0 = a_0 < a_1 < \dots < a_k = c$ be integers and define the coarsened response $Y_i^* = j$ when $a_{j-1} < Y_i \leq a_j$. Then Y_i^* follows the proportional odds model with the *same* β and cutpoints $\theta_j^* = \theta_{a_j}$, $j = 1, \dots, k-1$. In particular dichotomizing at any cut r gives a binary logistic model with slope β ;
- (c) (*coherence*) the ordering constraint Eq. 36.2.3 reduces to $\theta_1 < \dots < \theta_{c-1}$ and so holds at every $\mathbf{x}$ at once;
- (d) (*interpretation*) for any two regressor vectors and every r ,

$$\frac{\text{odds}(Y > r \mid \mathbf{x}_2)}{\text{odds}(Y > r \mid \mathbf{x}_1)} = \exp\{(\mathbf{x}_2 - \mathbf{x}_1)^\top \beta\}, \quad (36.3.2)$$

a single odds ratio that does not depend on r ; and if $(\mathbf{x}_2 - \mathbf{x}_1)^\top \beta > 0$ then the response at $\mathbf{x}_2$ is stochastically larger than the response at $\mathbf{x}_1$.

Proof.

- (a) If such a latent variable exists, then $P(Y_i \leq r) = P(U_i \leq \theta_r) = P(\varepsilon_i \leq \theta_r - \mathbf{x}_{(i)}^\top \beta) = \Lambda(\theta_r - \mathbf{x}_{(i)}^\top \beta)$, which is Eq. 36.3.1 after taking logits, since $\log\{\Lambda(z)/(1 - \Lambda(z))\} = z$. Conversely, given Eq. 36.3.1, define U_i by the displayed construction with ε_i standard logistic; the computation just made shows that the induced distribution of Y_i is the one specified by the model, and the cutpoints are increasing by (c).
- (b) The events are nested: $\{Y_i^* \leq j\} = \{Y_i \leq a_j\}$, directly from the definition. Hence $P(Y_i^* \leq j \mid \mathbf{x}_{(i)}) = \gamma_{i,a_j}$, and applying the logit to both sides, $\log\{\gamma_{i,a_j}/(1 - \gamma_{i,a_j})\} = \theta_{a_j} - \mathbf{x}_{(i)}^\top \beta$. This is Eq. 36.3.1 for Y^* with cutpoints θ_{a_j} , which are increasing because the θ_r are, and with β untouched. Taking $k = 2$, $a_1 = r$ gives the binary statement.
- (c) Substitute $\beta_r = \beta$ into Eq. 36.2.3: the terms $\mathbf{x}_{(i)}^\top \beta$ cancel, leaving $\theta_1 < \dots < \theta_{c-1}$, a condition on the parameters alone. This is the case already recorded in Proposition 36.2.1.

- (d) From Eq. 36.3.1, $\text{odds}(Y \leq r \mid \mathbf{x}) = \exp(\theta_r - \mathbf{x}^\top \boldsymbol{\beta})$, so $\text{odds}(Y > r \mid \mathbf{x}) = \exp(\mathbf{x}^\top \boldsymbol{\beta} - \theta_r)$ and the ratio is $\exp\{(\mathbf{x}_2 - \mathbf{x}_1)^\top \boldsymbol{\beta}\}$; the cutpoint cancels, which is the whole content of Eq. 36.3.2. If $(\mathbf{x}_2 - \mathbf{x}_1)^\top \boldsymbol{\beta} > 0$ the ratio exceeds one for every r , so $P(Y > r \mid \mathbf{x}_2) > P(Y > r \mid \mathbf{x}_1)$ for every r : stochastic dominance. ■

Idea. Part (b) is why the model is worth having: the slope means the same thing however finely the scale was cut, so studies using three categories and seven are comparable. It is also the model's exposed flank, since *every* dichotomization must give that slope.

Warning (What proportional odds does not say). Eq. 36.3.2 is about *cumulative* odds. It does not say that a unit of x has the same effect on every category probability: differencing $\pi_{ir} = G(\eta_{ir}) - G(\eta_{i,r-1})$ gives $\partial \pi_{ir} / \partial \mathbf{x} = \{g(\eta_{i,r-1}) - g(\eta_{ir})\} \boldsymbol{\beta}$, whose sign changes as r moves across the distribution. Nor does it say that the latent variable of part (a) exists in any physical sense: the construction is a *device for generating the model*, always available once the model holds, and it carries no empirical content. Claims that the latent scale is “really” attitude, utility or severity are claims about the world, and reading $\boldsymbol{\beta}$ causally needs the whole apparatus of Chapter 25.

36.3.1 Checking the assumption

The restriction is $(c - 2)p$ linear constraints on the unrestricted cumulative model, so the three statistics of Proposition 34.5.3 are all available. The likelihood ratio and Wald tests need the unrestricted model, which by Proposition 36.2.1 may not be fittable; the score test needs only the restricted fit, which always is.

Proposition 36.3.2 — Testing and diagnosing the proportional odds restriction

Let $A \subseteq \{1, \dots, p\}$ index the regressors whose slopes are allowed to vary, and embed Eq. 36.3.1 in the cumulative logit model $\text{logit } \gamma_{ir} = \theta_r - \mathbf{x}_{(i)}^\top (\boldsymbol{\beta} + \boldsymbol{\delta}_r)$, where $\boldsymbol{\delta}_1 = \mathbf{0}$ and $\boldsymbol{\delta}_r$ is supported on A for $r \geq 2$. Write $\boldsymbol{\psi} = (\boldsymbol{\theta}, \boldsymbol{\beta}, \boldsymbol{\delta})$ and let $\hat{\boldsymbol{\psi}}_0$ be the maximum likelihood estimate under $\boldsymbol{\delta} = \mathbf{0}$. Then, under the regularity conditions recorded in Section 36.2 and if Eq. 36.3.1 holds:

- (a) the score statistic $S_{\text{sc}} = \mathbf{u}(\hat{\boldsymbol{\psi}}_0)^\top \mathbf{J}(\hat{\boldsymbol{\psi}}_0)^{-1} \mathbf{u}(\hat{\boldsymbol{\psi}}_0)$, with $\mathbf{u}$ and $\mathbf{J}$ from Eq. 36.2.4, is computable from the restricted fit alone, equals $\mathbf{u}_\delta^\top [\mathbf{J}^{-1}]_{\delta\delta} \mathbf{u}_\delta$, and converges in distribution to χ^2 with $(c - 2)|A|$ degrees of freedom;
- (b) the likelihood ratio and Wald statistics for the same hypothesis have the same limit and are asymptotically equivalent to S_{sc} ;
- (c) the $c - 1$ separate binary logistic regressions of $\mathbf{1}\{Y_i \leq r\}$ on $\mathbf{x}_{(i)}$ estimate the same $\boldsymbol{\beta}$, so plotting their slopes with confidence intervals shows where any departure lies. The $c - 1$ fits are dependent, so the plot is a diagnostic and not a test.

Proof.

- (a) By definition of a maximum, the components of $\mathbf{u}(\hat{\boldsymbol{\psi}}_0)$ in the $(\boldsymbol{\theta}, \boldsymbol{\beta})$ block vanish: those are the likelihood equations of the restricted model, which is the full model with $\boldsymbol{\delta}$ held at zero. Writing $\mathbf{u} = (\mathbf{0}^\top, \mathbf{u}_\delta^\top)^\top$ and partitioning $\mathbf{J}^{-1}$ conformably gives the stated reduction. That $\mathbf{u}^\top \mathbf{J}^{-1} \mathbf{u}$ has the χ^2 limit with degrees of freedom the number of restrictions is the score-test half of Proposition 34.5.3, which for a generalized linear model is proved there and holds here by the same argument applied to the likelihood of Proposition 36.2.2, under the conditions imported in Section 36.2; the restrictions number $(c - 2)|A|$, one vector $\boldsymbol{\delta}_r$ of length $|A|$ for each of the cuts $r = 2, \dots, c - 1$.
- (b) The asymptotic equivalence of the three statistics is Proposition 34.5.3, and it carries over under the same imported conditions; the hypothesis is a linear restriction on a parameter interior to the parameter space.
- (c) Dichotomizing at cut r gives, by Theorem 36.3.1(b), the binary logistic model logit $P(Y \leq r) = \theta_r - \mathbf{x}^\top \boldsymbol{\beta}$, with intercept θ_r and the same $\boldsymbol{\beta}$. Each fit is therefore a correctly specified binary logistic regression under the null and its estimate is consistent for $\boldsymbol{\beta}$ by Theorem 34.5.1. The fits use overlapping data, so their errors are correlated, which is why they cannot simply be pooled or differenced into a test.

■

Brant (1990) turns (c) into a test by estimating the joint covariance of the $c - 1$ slope estimates — they solve stacked estimating equations, so a sandwich-type covariance in the spirit of Theorem 21.4.2 is available — and forming a Wald statistic for their equality. It tests the same null hypothesis and has the same chi-squared limit. It is *not* the score statistic, it is built from different estimating equations, and the two can differ appreciably in small samples.

Warning (The test is not the question). With n in the thousands the score test rejects departures too small to matter; with n in the dozens it misses departures that would change every conclusion. The plot in (c) answers the useful question — *how much* do the slopes move, and where. Report both.

36.3.2 Self-placement, checked

Example. (The proportional odds model does not hold)

The common odds ratio of Theorem 36.3.1(d) is worth having only if it is there. For Example (A seven-point scale), let all three slopes vary across the six cuts. The score statistic of Proposition 36.3.2 is 67.667 on 15 degrees of freedom, $p = 1.16 \times 10^{-8}$; the likelihood ratio statistic against the fitted unrestricted model is 67.420, agreeing to within half a percent as Proposition 36.3.2(b) predicts. One regressor at a time, on $c - 2 = 5$ degrees of freedom:

regressor	score statistic	p	likelihood ratio
age	17.542	3.58×10^{-3}	18.367
education	41.258	8.32×10^{-8}	40.190
income	20.056	1.22×10^{-3}	20.155

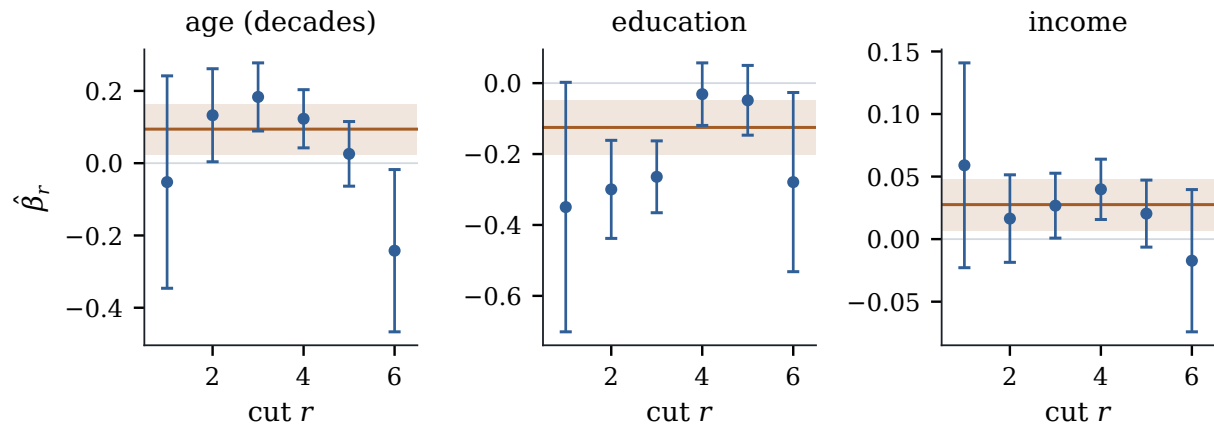


Figure 36.3.1: Slopes from the six separate binary logistic fits of $\mathbf{1}\{Y \leq r\}$ on the three regressors, with 95% intervals; the horizontal line and band are the proportional odds estimate and its interval. Under proportional odds the six points scatter about the line ([Proposition 36.3.2](#)). Education does not.

Education is the main offender. [Figure 36.3.1](#) shows why: its slope is near -0.3 at the three lowest cuts, near 0 at cuts 4 and 5, and back near -0.3 at cut 6. Education sharply separates the liberal end of the scale from everything above it, says almost nothing about position within the upper middle, and then separates the extreme conservative category again. That is a fact about the data, and a single number cannot express it.

The cells on this page continue those of [Section 36.2](#); this first one repeats the data and the fitting routine so that the page runs on its own.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

NAMES = ["age (decades)", "education", "income"]

anes = sm.datasets.anes96.load_pandas().data
y = anes["selfLR"].astype(int).to_numpy()          # 1 = extremely liberal ... 7 =
    extremely conservative
X = np.column_stack([anes["age"] / 10.0, anes["educ"], anes["income"]])
n, p = X.shape
c = 7
Y = np.zeros((n, c))
Y[np.arange(n), y - 1] = 1.0

def cum_gamma(psi, X, c, free):
    n, p = X.shape
    q, nf, fl = len(psi), len(free), list(free)
    theta, beta = psi[:c - 1], psi[c - 1:c - 1 + p]
    delta = psi[c - 1 + p:].reshape(c - 2, nf)
    G = np.empty((n, c - 1))
    dG = np.zeros((n, c - 1, q))
    for r in range(c - 1):
        beta_r = beta.copy()
        if r >= 1 and nf:
            beta_r[fl] = beta_r[fl] + delta[r - 1]
        g = 1.0 / (1.0 + np.exp(-(theta[r] - X @ beta_r)))
```

```

        lam = (g * (1.0 - g))[:, None]          # the logistic density at the
cut
        G[:, r] = g
        dG[:, r, r] = lam[:, 0]
        dG[:, r, c - 1:c - 1 + p] = -lam * X
        if r >= 1 and nf:
            off = c - 1 + p + (r - 1) * nf
            dG[:, r, off:off + nf] = -lam * X[:, fl]
    return G, dG

def cum_probs(psi, X, c, free):
    G, dG = cum_gamma(psi, X, c, free)
    P = np.diff(np.column_stack([np.zeros(len(X)), G, np.ones(len(X))]), axis=1)
    D = np.concatenate([dG[:, :1, :], np.diff(dG, axis=1)], axis=1)
    return P, D

def cum_score(psi, X, Y, c, free):
    P, D = cum_probs(psi, X, c, free)
    Pm = P[:, :c - 1]
    Sigma = np.einsum("ir,rs->irs", Pm, np.eye(c - 1)) - np.einsum("ir,is->irs", Pm,
        Pm)
    A = np.linalg.solve(Sigma, D)              #  $\Sigma_i^{-1} D_i$ , one
        observation at a time
    U = np.einsum("irk,ir->k", A, Y[:, :c - 1] - Pm)
    J = np.einsum("irk,irl->kl", D, A)
    return U, J

def cum_loglik(psi, X, Y, c, free):
    P, _ = cum_probs(psi, X, c, free)
    return float(np.sum(Y * np.log(np.clip(P, 1e-300, None))))

def cum_fit(X, Y, c, free=(), tol=1e-10, maxit=100):
    n, p = X.shape
    psi = np.zeros((c - 1) + p + (c - 2) * len(free))
    psi[:c - 1] = np.linspace(-1.5, 1.5, c - 1)    # start from ordered cutpoints,
        no slopes
    for it in range(1, maxit + 1):
        U, J = cum_score(psi, X, Y, c, free)
        step = np.linalg.solve(J, U)
        psi = psi + step
        if np.max(np.abs(step)) < tol:
            break
    return psi, J, it

psi, J, iters = cum_fit(X, Y, c)
theta, beta = psi[:c - 1], psi[c - 1:]
se = np.sqrt(np.diag(np.linalg.inv(J)))
ll_po = cum_loglik(psi, X, Y, c, ())
print(f"log-likelihood {ll_po:.4f} after {iters} Fisher scoring steps")

```

[Run this code online](#)

```

# the same slopes estimated one cut at a time: c - 1 separate binary logistic fits

```

```

Xc = sm.add_constant(X)
cut_beta = np.empty((c - 1, p))
cut_se = np.empty((c - 1, p))
for r in range(1, c):
    fit_r = sm.Logit((y <= r).astype(float), Xc).fit(disp=0)
    cut_beta[r - 1] = -np.asarray(fit_r.params)[1:]      # logit P(Y <= r) = theta_r -
    x^T beta_r
    cut_se[r - 1] = np.asarray(fit_r.bse)[1:]

for r in range(c - 1):
    print(f"cut {r + 1}: " + " ".join(f"{b:7.4f} ({s:.4f})" for b, s in zip(
        cut_beta[r], cut_se[r])))

```

[Run this code online](#)

The score test needs no second fit: evaluate the score and information of the *larger* model at the restricted estimate, with the extra coefficients zero.

```

# score test of proportional odds: all three slopes free to vary across cuts
free = (0, 1, 2)
psi_free = np.zeros((c - 1) + p + (c - 2) * len(free))
psi_free[:c - 1 + p] = psi                                # the proportional odds fit,
    with delta = 0
U, J_free = cum_score(psi_free, X, Y, c, free)
score_stat = float(U @ np.linalg.solve(J_free, U))
df = (c - 2) * len(free)

psi_np, _, _ = cum_fit(X, Y, c, free=free)
ll_np = cum_loglik(psi_np, X, Y, c, free)
lr_stat = 2 * (ll_np - ll_po)

print(f"score {score_stat:.3f} on {df} df, p = {stats.chi2.sf(score_stat, df):.3g}")
print(f"LR      {lr_stat:.3f} on {df} df, p = {stats.chi2.sf(lr_stat, df):.3g}")

```

[Run this code online](#)

36.3.3 When it fails

Rejecting proportional odds is information, not a disaster. The options, roughly in order of how much they give up.

Partial proportional odds. Free the slopes of the offending regressors only, as in [Proposition 36.3.2](#), keeping the restriction for the rest: the model of Peterson and Harrell (1990). It costs the coherence guarantee, so Proposition 36.2.1 must now be checked at every $\mathbf{x}$ of interest. Freeing education alone raises the log-likelihood to -1593.7610 , a likelihood ratio statistic of 40.190 on five degrees of freedom ($p = 1.37 \times 10^{-7}$), and lowers Akaike's criterion (Definition 29.2.2) from 3245.71 to 3215.52; the fully unrestricted model reaches 3208.29 with ten more parameters. The fitted education slopes by cut are -0.2683 , -0.2988 , -0.2523 , -0.0206 , -0.0498 and -0.2970 , and every fitted probability stays positive.

```

# partial proportional odds: education alone gets cut-specific slopes
psi_pp, J_pp, _ = cum_fit(X, Y, c, free=(1,))
ll_pp = cum_loglik(psi_pp, X, Y, c, (1,))
educ_slopes = np.concatenate([[psi_pp[c - 1 + 1]], psi_pp[c - 1 + 1] + psi_pp[c - 1
    + p:]])

```

```

aic = lambda ll, k: -2 * ll + 2 * k
print("education slope by cut:", np.round(educ_slopes, 4))
print("AIC proportional odds %.2f | partial %.2f | unrestricted %.2f"
      % (aic(ll_po, c - 1 + p), aic(ll_pp, len(psi_pp)), aic(ll_np, len(psi_np))))

```

[Run this code online](#)

A different link. Non-proportionality is sometimes link misspecification in disguise: the three links disagree in the tails (Proposition 35.2.1), so data can satisfy a cumulative probit or complementary log-log model with common slopes while violating the logit version. Trying the other two costs nothing.

A scale effect. Let the latent variable of Theorem 36.3.1(a) have a dispersion that depends on the regressors, $\log \sigma(\mathbf{x}) = \mathbf{x}^\top \boldsymbol{\gamma}$. The cumulative logits become $(\theta_r - \mathbf{x}^\top \boldsymbol{\beta})e^{-\mathbf{x}^\top \boldsymbol{\gamma}}$, non-parallel in a particular way: they fan out rather than crossing arbitrarily (Exercise (B2)). This costs p parameters instead of $(c - 2)p$, and when it fits, “the effect is the same but the responses are more spread out here” is a far more useful statement than a table of slopes.

A different family. Data that violate proportional odds badly may satisfy a sequential model (Section 36.4) with common slopes; failing that, the baseline-category model abandons the ordering but is never wrong about it.

Report the cuts. Sometimes no single number describes the effect, and the honest answer is Proposition 36.3.2(c). What is *not* a remedy is keeping the proportional odds fit and repairing its standard errors with a sandwich estimator: Theorem 21.4.2 fixes the variance of an estimator whose target is still well defined, and here the target itself has evaporated.

36.3.4 Exercises

36.3.4.1 A. Check your understanding

Exercise A1.

A proportional odds model has $c = 4$ and $p = 3$. How many parameters does it have? How many does the model with cut-specific slopes have, and what are the degrees of freedom of the score test of Proposition 36.3.2 with $A = \{1, 2, 3\}$ and with $A = \{2\}$?

Exercise A2.

In a proportional odds fit the coefficient of a treatment indicator is $\hat{\beta} = 0.7$. Write down three correct sentences and one common incorrect one.

Solution. Correct: the odds that a treated subject exceeds any fixed level are $e^{0.7} \approx 2$ times the odds for a control; the same factor applies at every level; the treated response is stochastically larger. Incorrect: “treatment doubles the probability of each category” — the effect on a category probability is a difference of two logistic densities times $\hat{\beta}$, and changes sign across the scale.

36.3.4.2 B. Practice

Exercise B1.

Using [Theorem 36.3.1\(b\)](#), predict what happens to the estimated slopes when the seven categories of Example (A seven-point scale) are collapsed to three, and compare with the actual refit in Exercise (B3). Does the comparison support or undermine the proportional odds assumption here?

Solution. Under the model the collapsed fit estimates the same β , so the two should differ only by sampling error, the collapsed one being less precise. The education slope moves from -0.1249 to -0.1148 , small against its standard error of 0.0416 , and the age slope from 0.0942 to 0.1350 , about one standard error: mild evidence at worst. Collapsing to three categories hides the failure that the cut-by-cut plot makes obvious, so a check should use all the cuts.

Exercise B2.

Derive the cumulative logits of the scale-effect model $U_i = \mathbf{x}_{(i)}^\top \beta + \sigma(\mathbf{x}_{(i)})\varepsilon_i$, $\log \sigma(\mathbf{x}) = \mathbf{x}^\top \gamma$, with ε_i standard logistic. Show that it contains [Eq. 36.3.1](#) as the case $\gamma = \mathbf{0}$, and that its cumulative logits never cross as functions of $\mathbf{x}$.

Solution. $P(Y_i \leq r) = P(\varepsilon_i \leq (\theta_r - \mathbf{x}^\top \beta)e^{-\mathbf{x}^\top \gamma})$, so $\text{logit } \gamma_{ir} = (\theta_r - \mathbf{x}^\top \beta)e^{-\mathbf{x}^\top \gamma}$, which is [Eq. 36.3.1](#) when $\gamma = \mathbf{0}$. For two cuts $r < s$ the difference of the cumulative logits is $(\theta_r - \theta_s)e^{-\mathbf{x}^\top \gamma}$, which has the sign of $\theta_r - \theta_s$ for every $\mathbf{x}$, since the exponential is positive: the curves fan out but never cross, so the ordering constraint of Proposition 36.2.1 holds everywhere.

36.3.4.3 C. Going deeper**Exercise C1.**

Show that if the proportional odds model holds and $\beta_j > 0$, then $E\{h(Y) \mid \mathbf{x}\}$ is increasing in x_j for *every* nondecreasing function h . Deduce that the sign of β_j is a statement that survives any recoding of the categories into numbers, as long as the recoding respects the order.

Solution. By [Theorem 36.3.1\(d\)](#), increasing x_j makes the response stochastically larger. Stochastic dominance is equivalent to $E\{h(Y) \mid \mathbf{x}_2\} \geq E\{h(Y) \mid \mathbf{x}_1\}$ for every nondecreasing h : write $E h(Y) = h(1) + \sum_{r=1}^{c-1} \{h(r+1) - h(r)\}P(Y > r)$, a sum of nonnegative coefficients times the survivor probabilities, each of which increases. Since any order-respecting scoring of the categories is such an h , the sign of β_j cannot be reversed by rescoreing.

Exercise C2.

Let [Theorem 36.3.1](#) hold with a single binary regressor, let Y_1 and Y_2 be independent responses from the two groups, and let U_1, U_2 be the latent variables of part (a) of that theorem.

(a) Show that $P(U_2 > U_1) = E\{\Lambda(\beta - \varepsilon)\}$ with ε standard logistic, so that it is a

function of β alone and strictly increasing in it.

- (b) Show that the concordance probability of the *observed* responses, $P(Y_2 > Y_1) + \frac{1}{2}P(Y_2 = Y_1)$, is increasing in β for fixed cutpoints. (Hint: write it as $E\{h(Y_2)\}$ for a nondecreasing h that depends only on the distribution of Y_1 .) Then evaluate it at $\beta = 1$ for $c = 3$ with cutpoints $(-0.5, 0.5)$ and again with $(-3, 3)$, and conclude that, unlike the odds ratio e^β , it is not free of the cutpoints.
- (c) The sample version of (b) is the Mann–Whitney statistic divided by $n_1 n_2$, with ties counted half. Say in what sense the proportional odds model is a regression version of the Wilcoxon–Mann–Whitney test, and in what sense it is not.

Solution.

- (a) $U_2 - U_1 = \beta + \varepsilon_2 - \varepsilon_1$, so $P(U_2 > U_1) = P(\varepsilon_1 < \beta + \varepsilon_2) = E\{\Lambda(\beta + \varepsilon_2)\}$, which is $E\{\Lambda(\beta - \varepsilon)\}$ by symmetry of the logistic law. No cutpoint appears, and Λ is strictly increasing, so this increases strictly with β ; at $\beta = 1$ it is 0.6613.
- (b) Conditioning on Y_1 gives $E\{h(Y_2)\}$ with $h(b) = P(Y_1 < b) + \frac{1}{2}P(Y_1 = b)$, nondecreasing in b and depending only on the distribution of Y_1 — which does not involve β , group one being the reference. Raising β makes Y_2 stochastically larger by [Theorem 36.3.1\(d\)](#), and $E\{h(Y_2)\}$ then increases, by the equivalence used in [Exercise \(C1\)](#). It is not a function of β alone: at $\beta = 1$ the cutpoints $(-0.5, 0.5)$ give 0.6370 and $(-3, 3)$ give 0.5482, while e^β is the same in both. Widely spaced cutpoints push mass into the middle category, so ties are common and the concordance probability is pulled towards $\frac{1}{2}$.
- (c) It is a regression version in that the two-group comparison is the same stochastic-ordering comparison, extended to several regressors and to adjustment. It is not, in two ways: the rank test is distribution free where the model imposes proportional odds at every cut, a real restriction by [Example \(The proportional odds model does not hold\)](#); and what the model reports is β , not the concordance probability, which by (b) cannot be recovered from β without the cutpoints.